Suyoung Lee

Curriculum Vitae

Contact Information

Affiliation:	Department of ECE, ASRI, Seoul National University, Seoul, Korea
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Research Interests

My research focuses on deep learning for computer vision, with a particular emphasis on 3D scene understanding and generation. I am currently working on accurate and high-quality 3D reconstruction and large-scale scene generation, with a growing interest in developing scalable 3D scene generation frameworks. In particular, I focus on holistic scene reconstruction from omnidirectional images. My earlier work in low-level vision—including video restoration and video frame interpolation—continues to inform my approach to producing high-fidelity visual outputs.

Education

Seoul National University (SNU), Seoul, Korea Mar. 2019 - Aug. 2026
Integrated M.Sc./Ph.D. course in Electric and Computer Engineering
GPA: 4.27 / 4.30 (60 credits in Ph.D. Course of Electrical and Computer Engineering)
Advisor: Prof. Kyoung Mu Lee

Seoul National University (SNU), Seoul, Korea Mar. 2013 - Feb. 2019
B.Sc. in Electric and Computer Engineering
Summa cum laude, GPA: 4.18/4.30 (136 credits in Electrical and Computer Engineering)
Leave for military service (Korea Airforce): Mar. 2015 - Mar. 2017 (24 months)

Gyeonggi Science High School, Suwon, Korea Mar. 2010 - Feb. 2013

Publications

- **Suyoung Lee** and Myungsub Choi, "Beyond the Verdict: Evidence-Aligned Evaluation of Visual Prompt-Injection Guardrails" in Proceedings of European Conference on Computer Vision (**ECCV**) Workshop **Oral Presentation**, 2026.
  2nd Workshop on Benchmarking Evidence-Aligned Multimodal Reasoning.

- **Suyoung Lee**, Jaeyoung Chung, and Kyoung Mu Lee, "OmniAdapt: Fast Test-Time Adaptation of Feed-Forward 3D Gaussian Splatting from Omnidirectional Images" (Under review).

- Jaeyoung Chung*, **Suyoung Lee***, Jianfeng Xiang, Jialong Yang, and Kyoung Mu Lee, "Map2World: Segment Map Conditioned Text to 3D World Generation," in Proceedings of European Conference on Computer Vision (**ECCV**), 2026.[1]
  works done during internship at Microsoft Research

- Sangmin Hong*, **Suyoung Lee***, and Kyoung Mu Lee, "StarryGazer: Leveraging Monocular Depth Estimation Models for Domain-Agnostic Single Depth Image Completion," to appear in Signal, Image and Video Processing (**SIVP**), Springer, 2026.

- Jaeyoung Chung*, **Suyoung Lee***, Hyeongjin Nam, Jaerin Lee, and Kyoung Mu Lee, "LucidDreamer: Domain-free Generation of 3D Gaussian Splatting Scenes," in IEEE Transactions on Visualization and Computer Graphics (**TVCG**) **Runner-Up of the 2025 Best Paper Award in TVCG Computer Graphics Area**, 2025 (300+ Citations, 1,500+ GitHub Stars).

- **Suyoung Lee***, Jaeyoung Chung*, Kihoon Kim, Jaeyoo Huh, Gunhee Lee, Minsoo Lee, and Kyoung Mu Lee, "OmniSplat: Taming Feed-Forward 3D Gaussian Splatting for Omnidirectional Images with Editable Capabilities," in Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (**CVPR**) **Highlight Presentation**, 2025.

- **Suyoung Lee***, Jaeyoung Chung*, Jaeyoo Huh, and Kyoung Mu Lee, "ODGS: 3D Scene Reconstruction from Omnidirectional Images with 3D Gaussian Splatting," in Proceedings of Neural Information Processing Systems (**NeurIPS**), 2024.

- Myungsub Choi, **Suyoung Lee**, Heewon Kim, and Kyoung Mu Lee, "Motion-aware dynamic architecture for efficient frame interpolation," in Proceedings of the IEEE/CVF International Conference on Computer Vision (**ICCV**), 2021.

- Sanghyun Son, **Suyoung Lee**, Seungjun Nah, Radu Timofte, and Kyoung Mu Lee, "NTIRE 2021 challenge on video super-resolution," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) Workshop, 2021.

- Seungjun Nah, Sanghyun Son, **Suyoung Lee**, Radu Timofte, and Kyoung Mu Lee, "NTIRE 2021 challenge on image deblurring," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) Workshop, 2021.

- **Suyoung Lee***, Myungsub Choi*, and Kyoung Mu Lee, "DynaVSR: Dynamic Adaptive Blind Video Super-Resolution," in Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), 2021.

Experience

• **Research Internship @ Microsoft Research Asia (Beijing)** Jul. 2025 - Dec. 2025
Topic: Large-scale 3D World Generation.
Group: Visual Computing Group.
Mentor: Jialong Yang `https://jlyang.org/`.
Role description: Adapted a pretrained transformer-based 3D foundation model (TRELLIS) to scene-level generation through a parameter-efficient adapter, working alongside the model's original authors.

[1] * indicates equal contribution

• **Industry Collaboration Project with LG AI Research** Nov. 2023 - Oct. 2024
Topic: Development of 3D Scene Reconstruction and Generation Technique for Real-world Applications.

• **Industry Collaboration Project with Mechatronics R&D Center, Samsung Electronics (Current Seminconductor R&D Center and AI Center)** Mar. 2020 - Dec. 2020
Topic: Learning-Based Data Interpolation Algorithm Using Multi-Layer Multi-Wavelength Inspection Images for Semiconductor Structure Analysis.

Service

• **Workshop Challenge Co-organizer @ CVPR 2021** Jun. 2021
NTIRE 2021 Challenge on Image Deblurring and Video Super-Resolution.

• **Reviewer**
IEEE TIP (2025-), IEEE TVCG (2026-).
CVPR (2023-), NeurIPS (2025-), ICCV (2023-), ECCV (2024-), WACV (2023).

Scholarships

• **Electrical Engineering and Computer Science Graduate Student Program**, 2019-2023
Korea Foundation for Advanced Studies

• **National Scholarship for Science & Engineering** 2013-2018
Korea Student Aid Foundation

Skills

Programming: PyTorch, Python (High), C++, CUDA, MATLAB (Low)
Language: Korean (Native), English (Fluent)

References

Advisor Kyoung Mu Lee
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`https://cv.snu.ac.kr/index.php/kmlee`